

Quantum Tunneling in Non-Standard Potentials

Progress Report

I have used opus 4.6 ai and this is 2 days of works.

What I've Done So Far

I built a working Schrodinger equation solver from scratch in Python. It doesn't rely on big scientific libraries, so it runs anywhere. I checked it against known solutions (infinite square well and harmonic oscillator) and the errors are less than 0.04%.

The Shape Factor Discovery

I studied six different barrier shapes (rectangular, Gaussian, triangular, parabolic, trapezoidal, exponential) with the exact same height and width.

My biggest finding so far is that the shape of the barrier completely changes the tunneling probability!

- * The exponential barrier lets the most particles tunnel through.
- * The rectangular barrier blocks the most.
- * This difference gets way bigger as the barriers get wider.

To measure this, I created a Shape Factor (S), which compares how much a specific shape blocks tunneling compared to a plain rectangular barrier.

What I'm Working On Next

- * Morse Potential: Looking at molecular vibrational levels, which connects this physics project to chemistry.
- * Double Well Tunnel Splitting: Studying how energy levels split when there are two wells next to each other.
- * Getting really clean data for my final paper.

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My Ultimate Aim

The ultimate goal of this project is to publish a computational quantum mechanics research paper! By systematically comparing how non-standard barrier shapes affect quantum tunneling (something textbooks usually ignore), I'm putting together a unique computational study. Once the data is ready, I plan to write it up and submit it to student research journals, arXiv, or science fairs like IRIS.